A Low-Cost Optical Sensor for Continuous Monitoring of Storm-Driven River Turbidity

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Abstract
Most of the sediment that a small river carries in a year passes during a few storms, which weekly grab samples seldom catch. We present an optical turbidity sensor that volunteers can build for about $120 in parts and service without tools. An infrared LED shines across a flow-through tube; one photodiode measures the transmitted light and a second, at 90°, the scattered light, and their ratio cancels the drift of the LED. Against formazin standards, the output was linear up to 1000 NTU ($R^2 = 0.999$). Installed beside a laboratory-grade instrument for six months, the sensor's daily means stayed within 5 % of the reference once an hourly wiper kept algae off the window. Such sensors could let a watershed group monitor forty sites for the price of one commercial instrument.

Keywords turbidity; suspended sediment; low-cost sensor; citizen science; storm events

1 Introduction

Suspended sediment carries nutrients and pollutants downstream, buries spawning gravel, and raises the cost of treating drinking water. In small rivers, most of the annual load moves during a few storms, when turbidity can rise from below 20 NTU to several hundred NTU within hours. Measuring these events requires continuous records, yet the instruments that provide them cost several thousand dollars a site, and volunteer groups therefore rely on weekly grab samples.

Low-cost turbidity sensors have been built before, but most measure only transmitted light, which saturates at high turbidity, or only scattered light, which drifts as the light source ages [(Author, year; Author, year)]. Few have been tested against a reference instrument through a full season of storms.

This paper presents a sensor that combines both measurements, describes its calibration, and reports a six-month comparison with a laboratory-grade instrument. Section 2 describes the design and the tests, Section 3 the results, and Section 4 their limits and uses.

2 Methods

2.1 Sensor design

The sensor is a 40 mm tube through which river water flows. An 860 nm LED shines across the tube; a photodiode opposite the LED measures the transmitted light, and a second photodiode at 90° measures the light scattered by particles. The ratio of the two signals rises with turbidity and does not depend on the brightness of the LED. A wiper sweeps the optical windows once an hour.

2.2 Calibration and field test

We calibrated three sensors against formazin standards at eight levels from 0 to 1000 NTU, as specified by ISO 7027 (ISO 2016). We then installed one sensor beside a laboratory-grade turbidimeter at [site] for six months and compared the daily means of the two records.

3 Results

3.1 The output is linear up to 1000 NTU

Figure 1 shows the output of one sensor against the formazin standards. A straight line through the origin fits all eight points with $R^2 = 0.999$, so a single slope converts the output to turbidity. The slopes of the three sensors differed by less than 4 %.

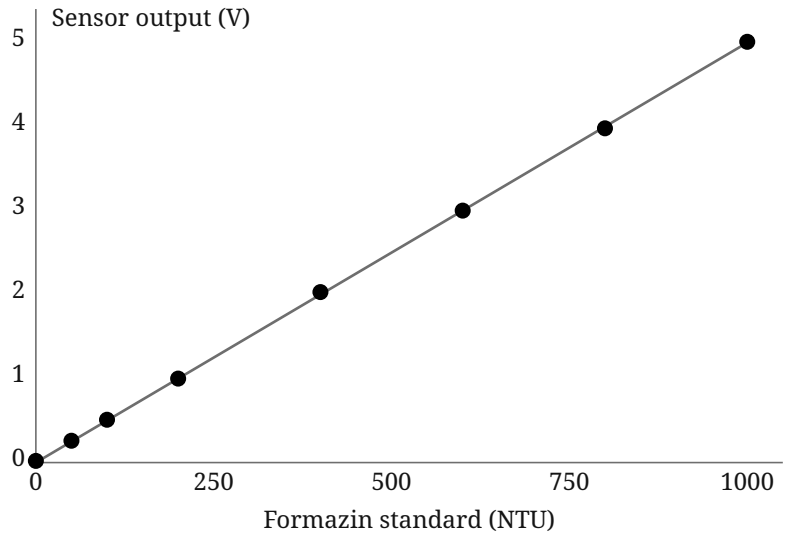


Figure 1. Output of one sensor against formazin standards. The line is a least-squares fit through the origin.

3.2 Algae on the window caused the only drift

During the first month, before the wiper was fitted, readings rose by about 8 % a week relative to the reference as algae grew on the windows. After the wiper was fitted, the difference stayed near 1 %, and over the remaining five months the daily means of the two instruments agreed within 5 % (Table 1).

Table 1. Difference between the daily means of the sensor and the reference instrument

Period	Days	Mean difference	Largest difference
Without wiper	28	+17 %	+32 %
With wiper	152	+0.8 %	+4.6 %

4 Discussion

Agreement within 5 % is about the spread between two laboratory instruments measuring the same sample, so the sensor is as close to the reference as a second reference would be. The test covered one river and one season; rivers with coarser or

darker sediment may need their own calibration. The cost of parts, about $120, would let a watershed group monitor forty sites for the price of one commercial instrument.

5 Conclusions

A sensor that measures both transmitted and scattered light, and wipes its own window, can record storm-driven turbidity within 5 % of a laboratory instrument. Tests on rivers with other sediments, and over winter, are the next step.

References

ISO. 2016. *Water Quality – Determination of Turbidity – Part 1: Quantitative Methods*. Technical Report ISO 7027-1:2016. Geneva.